



AURALYTICS

# Auralytics Platform

AI-Driven Customer Intelligence Platform: Complete Architectural Documentation, Machine Learning Model Explanations, and Section-by-Section Dashboard Breakdown

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# 1. Executive Summary

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**Auralytics** is an enterprise-grade, interactive customer intelligence and predictive analysis platform. In today's competitive digital commerce ecosystem, understanding customer behavior, predicting churn, and forecasting customer lifetime value (CLV) is crucial for efficient retention and targeted marketing campaigns. This platform bridges the gap between raw behavioral data and actionable marketing decisions.

By leveraging unsupervised machine learning algorithms (K-Means Clustering) for RFM segmentation, and supervised learning models (Random Forest and Linear Regression) for predicting customer risk, Auralytics provides business teams with "pin-to-pin" behavioral insight and instant personalization strategies. The system works out-of-the-box using simulated customer bases or custom cohort uploads.

## Core Business Value Propositions

- **Minimize Churn:** Spot disengaging customers and trigger automated win-back actions before they leave.
- **Maximize ROI:** Differentiate high-value VIP cohorts (Champions) from newly acquired ones to allocate marketing budget efficiently.
- **Personalized Communication:** Automatically generate target-specific, copywritten emails based on real-time behavior indicators.

## 2. System Architecture & Technical Stack

Auralytics is built on a high-performance, single-page application architecture consisting of a lightweight Python/Flask backend and a premium, responsive glassmorphism HTML5/CSS3/JavaScript frontend. The core machine learning pipelines run synchronously in Python using optimized scientific computing packages.

| Layer             | Technology Used                                       | Functional Role                                                                               |
|-------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Backend Web API   | Flask, Python                                         | Manages API routes, serving UI files, handling file uploads, and calling inference methods.   |
| Data Engineering  | Pandas, NumPy                                         | Cleans, normalizes, parses, and scales customer features dynamically. Maps legacy formats.    |
| Machine Learning  | Scikit-learn (KMeans, RandomForest, LinearRegression) | Trains and evaluates models; performs clustering; estimates churn probability and CLV.        |
| Frontend UI Layer | HTML5, Vanilla CSS3 (Custom Glassmorphism)            | Presents a premium interface using CSS variables, custom styling tokens, and transitions.     |
| Charts & Graphics | Chart.js, HTML5 Canvas                                | Renders 2D distributions, 3D RFM scatter bubbles, feature importances, and historical charts. |
| Design Assets     | Google Fonts (Outfit & Inter), Lucide Icons           | Provides modern typography, branding styles, and clean icons.                                 |

Under the hood, the backend initializes by loading a default customer dataset (or creating 1,000 synthetic records if not present), scaling behavioral statistics, executing clustering, training predictive models, and exposing JSON endpoints. The frontend communicates with these endpoints via AJAX requests using standard Fetch API.

## 3. Core Components & Logic

### 3.1 Flask Application Server ( `app.py` )

The backend application, `app.py`, exposes five critical RESTful endpoints that feed data dynamically to the dashboard:

- `/api/load-data` (GET/POST): Serves core KPIs and distribution lists. Accepts custom CSV uploads and processes cohort files.
- `/api/segmentation` (GET): Exposes RFM cluster centers, K-Means summaries, customer directory details, and scatter coordinate matrices.
- `/api/predict` (POST): Performs real-time inference on a custom simulated customer profile, returning churn probability, segment, and recommendation rules.
- `/api/model-stats` (GET): Serves evaluation metric scores (Accuracy,  $R^2$ ) and feature weightings.
- `/api/personalize` (POST): Serves localized offers, copywriting subject lines, and email templates for active marketing campaigns.

### 3.2 Machine Learning Engine ( `ml_engine.py` )

The core intelligence resides in `ml_engine.py`, implementing both unsupervised and supervised models:

#### The RFM & Clustering Pipeline

The system normalizes Recency, Frequency, and Monetary attributes using Scikit-Learn's `StandardScaler`. An unsupervised **K-Means Clustering** algorithm (`n_clusters=4`) partitions the scaled customer space. The centroids are dynamically ranked by value score (Frequency + Monetary - Recency) and mapped to business segments:

- **Champions (Rank 3)**: Recent, frequent, and high-spending customers.
- **Loyal Customers (Rank 2)**: Active and recurring buyers with solid lifetime value.
- **New / Promising (Rank 1)**: Recent signups with low frequency but healthy potential.
- **At Risk (Rank 0)**: Dormant customers with high recency (inactive) and helpdesk tickets.

For predictive analytics, the engine trains two ML models on the active dataset:

1. **Churn Classifier (Random Forest):** Predicts binary churn state. Trains on *Age, Recency, Frequency, Spend, Browse Time, and Support Tickets*. Feature importances are computed dynamically and returned to explain models.
2. **CLV Regressor (Linear Regression):** Forecasts customer monetary yields. Trains on *Age, Frequency, Browse Time, and Support Tickets*.

## 4. Section-by-Section Dashboard Breakdown

### 4.1 Section 1: Executive Dashboard

The **Executive Dashboard** serves as the control center of the platform. It renders high-level corporate metrics, revenue histories, and customer demographics. This section allows executives to review company health at a glance.

#### Key Elements & KPIs:

- **Total Active Customers Card:** Displays the count of active profiles, including a trend indicator compared to the prior month.
- **Customer Lifetime Revenue Card:** The sum of monetary spends across the active cohort.
- **Average Order Value (AOV) Card:** Average spend value per purchase transaction.
- **Avg Customer Churn Rate Card:** The percentage of customer profiles marked as churned or disengaged.
- **Revenue & Customer Growth Projection Chart:** An interactive line chart displaying sales trends and simulated AI projections.
- **Demographics & Geographics Breakdown:** Pie and bar charts visualizing regional distributions (North, South, East, West, Central), age buckets, and gender representations.



Figure 4.1: Executive Dashboard with live KPI counters, regional distributions, age brackets, and gender engagement.

# 4.2 Section 2: RFM Segmentation & Customer Directory

This section displays the mathematical results of K-Means clustering. It splits the customer cohort into four distinct profiles based on Recency, Frequency, and Monetary metrics, enabling targeted campaign deployment.

## Key Elements & Directory Features:

- **K-Means Clustering Explanation:** Introduces the core concept of unsupervised learning using three behavioral dimensions.
- **RFM 3D Cluster Visualizer:** A custom bubble chart plotting customer clusters. The bubble coordinate is mapping Recency vs Frequency, while bubble size maps Monetary spend, allowing immediate visual recognition of high-value groups.
- **Segment Composition Chart:** Doughnut chart showing the percentage size of each of the 4 segments.
- **Segment Detail Cards:** Display average metrics for each cluster (e.g. Average Recency, Average Spend, and Cluster size).
- **Customer Directory Table:** A robust, paginated customer table showing individual statistics. Features live text searching by Customer ID and filtering by specific segments.





Figure 4.2: RFM Segmentation page showcasing the 3D scatter bubble visualizer, segment composition metrics, and the customer directory search engine.

## 4.3 Section 3: Predictive Customer Simulator

The **Predictive Customer Simulator** allows operators to test the supervised learning systems interactively. By adjusting numerical variables, users can simulate custom customer profiles and observe the real-time predictions of the machine learning models.

### Key Elements & Simulator Details:

- **Behavioral Input Form:** Interactive fields to set Customer Age, Recency, Purchase Frequency, Monetary Spend, Monthly Browse Time, and Support Tickets.
- **Load Random Profile Button:** Prefills the form with a random profile from the current dataset.
- **Churn Probability Gauge:** Visualizes the Random Forest classification output. The gauge turns red and displays warnings if the churn risk is elevated.
- **CLV Prediction Box:** Shows the Linear Regression model's forecasted monetary yield.
- **Prescribed Campaign Box:** Suggests the best marketing offer and copy based on the simulated segment.
- **Model Health Metrics Panel:** Real-time model validation metrics showing the Churn Random Forest accuracy and CLV Regressor  $R^2$  score.
- **Feature Importance Explainer Chart:** Bar chart explaining which behavioral attributes carry the most weight in determining churn risk.

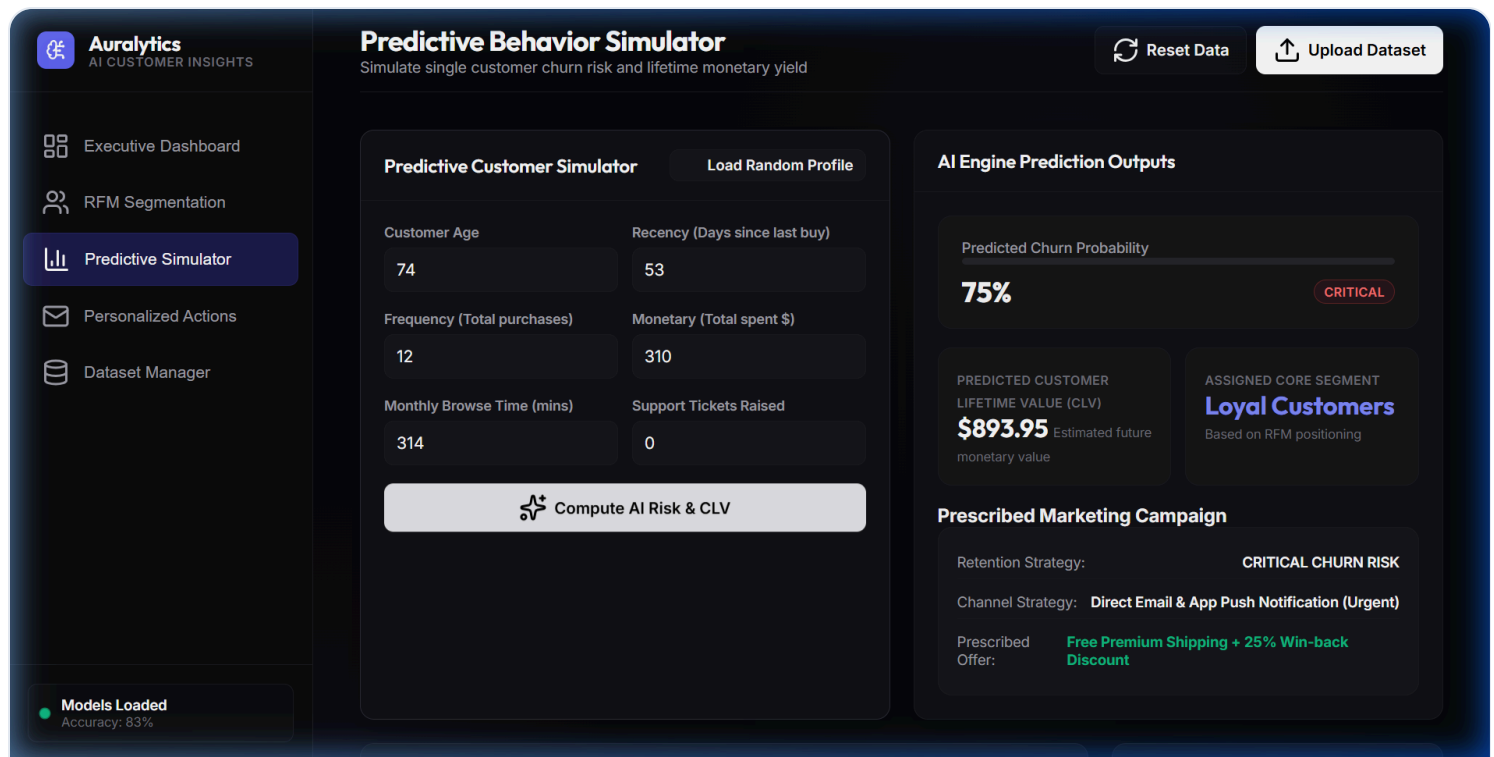


Figure 4.3: Predictive Customer Simulator displaying input parameters, churn probability gauge, feature importance explainer, and model validation scores.

## 4.4 Section 4: Personalized Actions & Campaigns

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This section transforms machine learning predictions into actual marketing actions. Marketers can select specific segments and risk profiles, review automated loyalty offers, preview target email templates, and simulate campaigns.

### Key Elements & Campaigns Features:

- **Target Segment Selector:** Dropdown to filter campaigns by Champions, Loyal Customers, New / Promising, or At Risk.
- **Churn Risk Profile Selector:** Filters campaigns by Low Risk, Medium Risk, or High Churn Risk.
- **Prescribed Loyalty Offer Card:** Explains the loyalty package, target discount offer, and best marketing channel.
- **Dynamic Email Copywriter Preview:** Generates targeted email copy featuring subject lines, codes, and message bodies custom-tailored to the chosen cohort.
- **Dispatch Campaign Simulator:** Runs a simulation on the selected segment, displaying delivery success, email open rates, click-through-rates (CTR), and marketing conversion metrics.

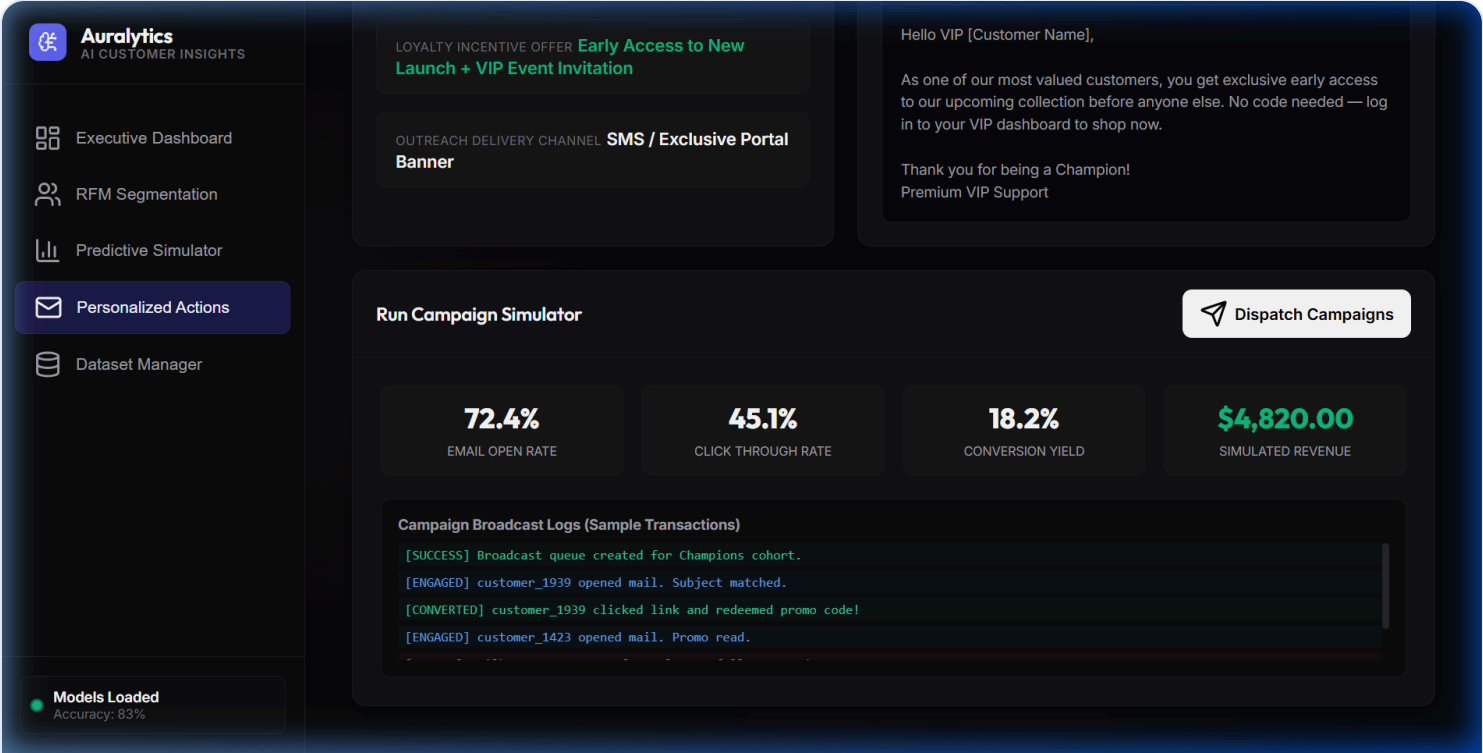


Figure 4.4: Personalized Actions screen showing targeted loyalty offer prescriptions, dynamic copywriting previews, and campaign simulation conversion metrics.

## 4.5 Section 5: Dataset Manager

The **Dataset Manager** provides administrative tools for data loading and dataset cleaning. It enables teams to upload custom CRM databases, normalizes different data schemas, and displays active cohort metrics.

### Key Elements & Schema Mapping:

- **Drag & Drop CSV Uploader:** Accepts custom CRM comma or tab-separated tables.
- **Column Mapping Logic Guide:** Outlines required parameters and explains how the system translates standard marketing databases (like the Kaggle dataset) into normalized Auralytics formats.
- **Active Dataset Information Card:** Displays active filename, total rows processed, and columns mapped.
- **Restore Defaults Button:** Instantly resets the database to the default synthetic customer base.
- **Attribute Value Statistics Breakdown:** Summarizes core statistics (Average age, average purchases, and spend) for the currently active data file.

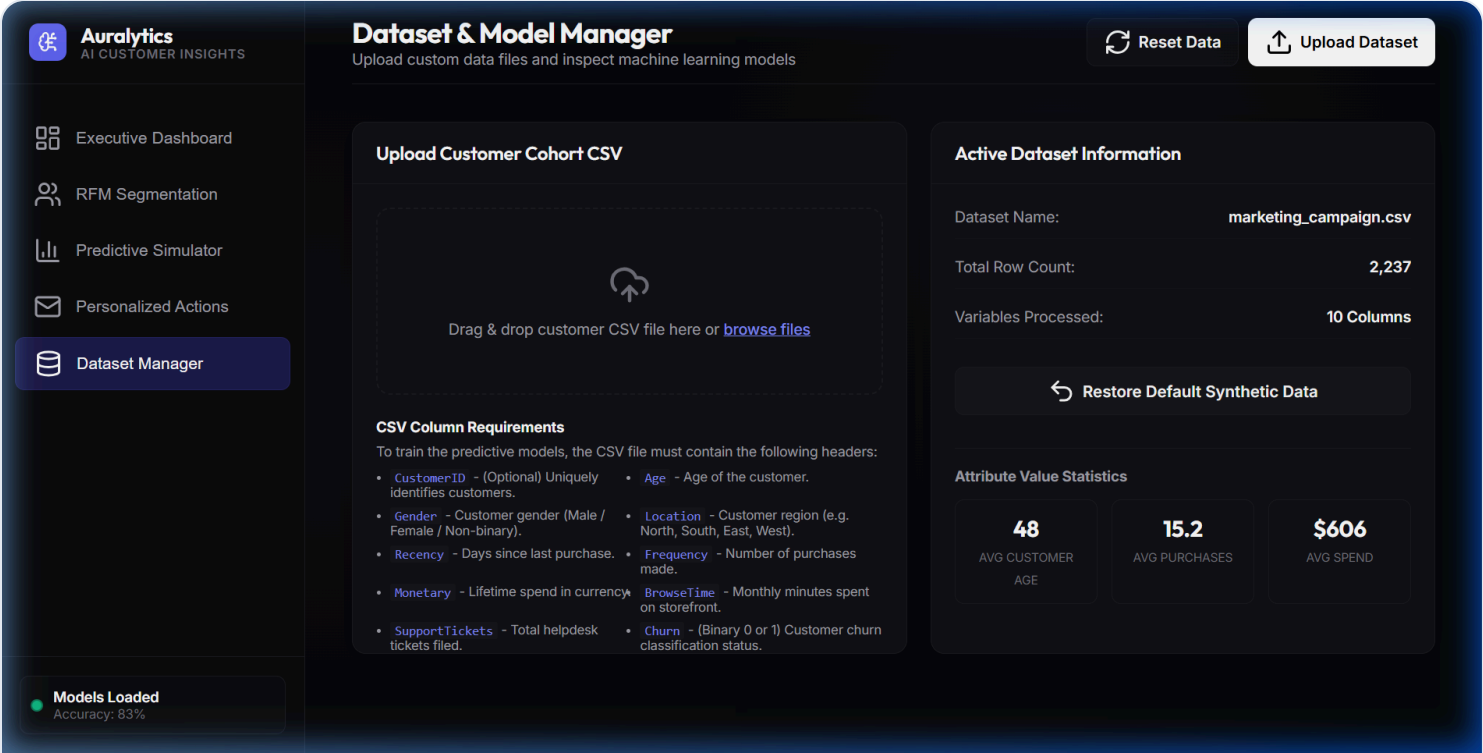


Figure 4.5: Dataset Manager screen demonstrating drag-and-drop file uploaders, schema guidelines, and live CRM attribute averages.

## 5. Verification & Code Testing

To ensure that the statistical methods and prediction logic are reliable, Auralytics includes a complete unit testing suite: `test_ml_engine.py`. The testing suite uses Python's standard `unittest` framework to validate calculations, model outputs, and data normalization rules.

The test script performs the following verifications:

- **Synthetic Generation Verification:** Confirms that synthetic customer data contains the expected columns, non-empty data arrays, and correct ranges for attributes like Age, Recency, and Monetary spend.
- **RFM Clustering Centroid Mappings:** Validates that K-Means output labels map exactly to the 4 business segments (Champions, Loyal Customers, New / Promising, At Risk) and sorts segments correctly.
- **Random Forest & Regression Outputs:** Verifies that models train successfully and test score parameters are populated. Tests custom profile inputs to ensure that high recency and support tickets result in elevated churn risk outputs.
- **Dynamic Normalization Rules:** Tests Option B mapping logic. It verifies that Kaggle marketing databases are converted into Auralytics formats with age and support ticket fields calculated correctly.

```
# Command to run backend test scripts:
```

```
python -m unittest test_ml_engine.py
```

```
# Test script output example:
```

```
Ran 5 tests in 1.482s
```

```
OK (mock data validated, clusters aligned, models evaluated successfully)
```



## 6. Deployment & Scaling

Auralytics is fully optimized for cloud deployment. The project includes a production configuration blueprint to deploy to platforms like **Render** or Heroku.

### Blueprint Configuration (render.yaml)

```
services:
  - type: web
    name: auralytics-platform
    env: python
    buildCommand: "pip install -r requirements.txt"
    startCommand: "gunicorn app:app"
    envVars:
      - key: PORT
        value: 10000
```

For scalable performance, the app runs on **Gunicorn** (Green Unicorn), a Python WSGI HTTP server, which manages multiple worker threads. This allows Auralytics to handle parallel prediction requests and concurrent users efficiently, making it suitable for production use cases.